

2(a)	I protective gear / education about transmission / contact tracing / early diagnosis / testing any two from: practising safe sex / example ; A protected sex / using protection during sex A barrier contraceptives I use contraceptives screening blood, donated for / before, transfusions AW or not taking blood from high risk individuals / AW ; <i>idea of treating donated blood (to inactivate viruses) ;</i> A example e.g. heat / chemical treatment solvent-detergent treatment <i>in context of drug abusers</i> avoid sharing, needles / syringes / drug injecting equipment AW or use, new / sterile, needles / syringes / drug injecting equipment ; A cookers / sterile containers to mix drugs A use needle-exchange schemes / AW A use safe injection sites / AW avoid breast feeding / use baby milk products instead of breast milk ; pre-exposure prophylaxis / PrEP or post-exposure prophylaxis / PEP ; AVP ; e.g. using sterile equipment for, tattooing / surgery / dental procedures / body piercing	2
2(b)	the number of times larger an image is than actual ; AW I or smaller A formula stated in words A symbolised I/A <i>to support a weakly worded answer</i> A (in context of describing an image) enlargement / making larger / making bigger / increase in size / AW	1

Question	Answer	Marks															
2(c)(i)	any one from: because bacteria have cell walls (only) of, murein / peptidoglycan ; because chitin is not found in bacterial cell walls ; A chitin / β-D-glucans, <u>main / major / AW</u> component of fungal cell walls	1															
2(c)(ii)	<table border="1"> <thead> <tr> <th>structural feature of <i>P. jirovecii</i></th><th>function</th><th>present (✓) or absent (x) in bacterial cells</th></tr> </thead> <tbody> <tr> <td>ribosomes</td><td>protein synthesis</td><td>✓</td></tr> <tr> <td>smooth endoplasmic reticulum</td><td>lipid / cholesterol / steroid, synthesis / metabolism AVP ; detoxification I carbohydrate metabolism</td><td>x</td></tr> <tr> <td>Golgi body</td><td>modification of proteins and lipids</td><td>x</td></tr> <tr> <td>mitochondrion / mitochondria</td><td>aerobic respiration</td><td>x</td></tr> </tbody> </table> one mark each correct column ;;; assume a tick that has been crossed is a cross ✕	structural feature of <i>P. jirovecii</i>	function	present (✓) or absent (x) in bacterial cells	ribosomes	protein synthesis	✓	smooth endoplasmic reticulum	lipid / cholesterol / steroid, synthesis / metabolism AVP ; detoxification I carbohydrate metabolism	x	Golgi body	modification of proteins and lipids	x	mitochondrion / mitochondria	aerobic respiration	x	3
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2(d)(i)	<i>allow squamous cells or squamous epithelial cells or epithelial cells for alveolar cells</i> <i>any two from:</i> 1 <i>gpA / (P. jirovecii) glycoprotein, <u>binds</u> to a, receptor / protein / glycoprotein on (surface of) alveolar cell</i> or <i>gpA <u>binds</u> to ECM, protein / glycoprotein ; I receptor</i> A <i>idea that ECM proteins are on, surface of / interacting with, (cell surface membrane of) alveolar cells</i> or <i>receptor / protein / glycoprotein on (surface of) alveolar cell is a binding site (for gpA) A same idea for ECM</i> 2 <i>(gpA) <u>complementary shape</u> ;</i> <i>allow ecf mp1 for gpA complementary shape to receptor on ECM</i> or <i>idea of complementary shape to attach / bind / adhere, to, alveolar cell / ECM</i> 3 <i>suggestion of adhering by, attractive charges / named bond types, between, gpA and, proteins / glycoproteins, on surface of alveolar cell / of ECM ;</i> 4 <i>AVP ; e.g. gpA acts as a <u>ligand</u></i>	2

Question	Answer	Marks
2(d)(ii)	any three from: 1 alveolar cells / ECM, hindered / surrounded by / AW, <i>P. jirovecii</i> cells ; A alveolar wall becomes, thicker / AW 2 <u>diffusion</u> (of oxygen), impaired / decreased / AW ; A distance for <u>diffusion</u> increased A reduced surface area for <u>diffusion</u> ; 3 between, alveolus / alveolar air / alveolar space, and, capillary / blood / red blood cell(s) ; <i>allow mp if oxygen is clearly going in the correct direction</i> 4 less / AW, oxyhaemoglobin formed ; AW e.g. A less oxygen, binds to / taken up by / associates with, haemoglobin A less oxygen forms bonds with, Fe²⁺ / iron ion / iron, (in haemoglobin) 5 <i>idea of</i> ability of elastic fibres to, stretch / recoil, impaired / AW ; A <i>idea of</i> less elastic / decreased elasticity (of alveoli) 6 alveolar air not, refreshed / AW, so decreased diffusion gradient ; 7 AVP ; e.g. presence of macrophages hinders diffusion of oxygen <i>suggestion that P. jirovecii</i> infection, damages / AW, alveolar capillaries / venules / branches of pulmonary vein <i>suggestion that</i> oxygen used up for <i>P. jirovecii</i>, metabolism / activity / AW less ability of collagen to provide support to alveoli / AW	3

Question	Answer	Marks
2(e)	any five from: casprofungin / drug, binds to site on, enzyme / glucan synthase, other than active site ; A binds to allosteric site changes shape of active site ; A enzyme changes, shape / tertiary structure, so active site is changed substrates cannot bind to active site / active site no longer complementary to substrates / enzyme-substrate complexes do not form ; A fewer enzyme-substrate complexes form less / no, (1,3- β-D) glucans / products, synthesised / produced ; cell wall, weakened / synthesis hindered / AW ; A cell wall not formed (leads to osmotic) lysis of, <i>P. jirovecii</i> / cell / fungus ; A bursting / cytolysis reduces / prevents, population growth ; AW e.g. decrease number of <i>P. jirovecii</i> A decreases ability of <i>P. jirovecii</i> to colonise / AW AVP ; e.g. <i>in context of lysis ref.</i> water entry into cell, by osmosis / down water potential gradient / AW (reduced number) increases chance of immune system eliminating the fungus	5